High-Dimensional Probability

Assignment #4

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Problem 1

Computing the VC dimension. Let the following concept classes be given:

(1) $\mathcal{T}$: all *filled* triangles in $\mathbb{R}^2$.

(2) $\mathcal{C}$: all *closed disks* in $\mathbb{R}^2$.

(3) $\mathcal{R}_n$: all axis-aligned hyperrectangles in $\mathbb{R}^n$.

Show that

$$\mathrm{VCdim}(\mathcal{T}) = 7, \qquad \mathrm{VCdim}(\mathcal{C}) = 3, \qquad \mathrm{VCdim}(\mathcal{R}_n) = 2n.$$

Solution

(1) Triangles. More generally, let $\mathcal{P}_d$ be the class of convex polygons in $\mathbb{R}^2$ with at most d sides. We show $\mathrm{VCdim}(\mathcal{P}_d) = 2d+1$. Then for triangles ($d = 3$),

$$\mathrm{VCdim}(\mathcal{T}) = 2 \cdot 3 + 1 = 7.$$

Lower bound. Place $2d+1$ points on a circle in cyclic order. Given any labeling, group the -1 labels into maximal consecutive blocks along the circle. Let the number of such blocks be b. Since labels alternate between -1 blocks and $+1$ blocks, having $b \geq d+1$ would force at least $2(d+1)$ points, impossible; hence $b \leq d$. For each negative block, draw the chord joining its endpoints, and take the halfplane that contains all the $+1$ points while excluding that arc of negative points. Intersect these $b \leq d$ halfplanes; the result is a convex polygon with at most d sides containing exactly the $+1$ points. Thus $2d+1$ points are shattered, so $\mathrm{VCdim}(\mathcal{P}_d) \geq 2d+1$. *Upper bound.* Take any $2d+2$ points in convex position and label them alternately $+1, -1, +1, -1, \ldots$ around the convex hull. Any convex polygon realizing this labeling must cross the hull boundary at least $2d+2$ times (each adjacent pair with different labels forces a boundary crossing along that edge). But a convex polygon with at most d sides can intersect the boundary of another convex polygon at most $2d$ times (each side contributes at most two intersection points after a generic perturbation). Contradiction. Hence $\mathrm{VCdim}(\mathcal{P}_d) \leq 2d+1$.

(2) Disks in $\mathbb{R}^2$. More generally, let $\mathcal{B}_n$ be the class of all closed Euclidean balls in $\mathbb{R}^n$. We show $\mathrm{VCdim}(\mathcal{B}_n) = n+1$, so for $n = 2$ we get $\mathrm{VCdim}(\mathcal{C}) = 3$. *Lower bound.* Pick linearly independent $a_1, \ldots, a_n \in \mathbb{R}^n$ and set $a_{n+1} = 0$. Given any labels $y_i \in \{\pm 1\}$, choose x solving

$$-2\langle x, a_i \rangle = -\|a_i\|_2^2 + y_i, \qquad i = 1, \ldots, n,$$

(which has a unique solution). Then pick r so that $\|x\|_2^2 - r^2$ has the sign needed to classify 0 as labeled. With a small margin choice, all inequalities match the intended labels. Hence $\mathrm{VCdim}(\mathcal{B}_n) \geq n+1$. *Upper bound.* Rewrite the membership condition as

$$\|a - x\|_2^2 \leq r^2 \quad \Longleftrightarrow \quad \langle(-2a, 1), (x, \|x\|_2^2 - r^2)\rangle + \|a\|_2^2 \leq 0,$$

which is an affine halfspace test in $\mathbb{R}^{n+1}$. If $n+2$ points in $\mathbb{R}^n$ were shattered by balls, then $n+2$ affine hyperplanes in $\mathbb{R}^{n+1}$ would realize all 2^{n+2} sign patterns. But m affine hyperplanes in $\mathbb{R}^d$ induce at most $\sum_{k=0}^{d} \binom{m}{k}$ sign patterns; for $(m, d) = (n+2, n+1)$ this equals $2^{n+2} - 1$. Contradiction. Hence $\mathrm{VCdim}(\mathcal{B}_n) \leq n+1$.

(3) Axis-aligned rectangles in $\mathbb{R}^n$. *Lower bound.* Let $S = \{\pm e_1, \ldots, \pm e_n\}$. Given any labeling of these $2n$ points, define intervals I_i coordinatewise to include/exclude $+e_i$ and $-e_i$ as desired (choose endpoints, e.g., from $\{-2, -\frac{1}{2}, \frac{1}{2}, 2\}$). Then $R = I_1 \times \cdots \times I_n$ realizes the labeling. So $\mathrm{VCdim}(\mathcal{R}_n) \geq 2n$. *Upper bound.* Let $X \subset \mathbb{R}^n$ with $|X| = 2n+1$. For each coordinate i, pick $u_i \in X$ minimizing and $v_i \in X$ maximizing the ith coordinate. Among $\{u_i, v_i\}_{i=1}^n$ there are at most $2n$ points, so choose $w \in X$ outside this set. Label all u_i, v_i as $+1$ and label w as -1. Any axis-aligned rectangle containing all u_i, v_i must contain the full coordinate-wise bounding box, hence must also contain w, contradiction. Therefore $\mathrm{VCdim}(\mathcal{R}_n) \leq 2n$.

Combining bounds gives

$$\mathrm{VCdim}(\mathcal{T}) = 7, \qquad \mathrm{VCdim}(\mathcal{C}) = 3, \qquad \mathrm{VCdim}(\mathcal{R}_n) = 2n.$$

Problem 2

Statistical Learning of Real-Valued Functions. Let the output space be the bounded interval $\mathcal{Y} = [-1, 1]$, and let $\mathcal{X}$ be an arbitrary input space. Let $\mathcal{F} \subseteq \{f : \mathcal{X} \to \mathcal{Y}\}$ be a class of real-valued functions. Assume we are given i.i.d. samples $\{(x_i, y_i)\}_{i=1}^n \sim P_{\mathcal{X}\mathcal{Y}}$, drawn from an unknown distribution on $\mathcal{X} \times \mathcal{Y}$. For a sample $x^n = (x_1, \ldots, x_n)$ and $1 \le p \le \infty$, let $\mathcal{N}_p(\mathcal{F}, \alpha, x^n)$ denote the empirical ℓ_p covering number.

1. **Covering numbers and metric monotonicity.** Let $1 \le p \le q \le \infty$. Show that for any sample $x^n = (x_1, \ldots, x_n)$,

$$\mathcal{N}_p(\mathcal{F}, \alpha, x^n) \;\le\; \mathcal{N}_q(\mathcal{F}, \alpha, x^n).$$

2. **Rademacher complexity via ℓ_1-covering.** Show that for any sample $x^n = (x_1, \ldots, x_n)$, the empirical Rademacher complexity satisfies

$$\widehat{\mathfrak{R}}(\mathcal{F}) \;\le\; \inf_{\alpha \ge 0} \Big(\alpha + \sqrt{\frac{2 \log \mathcal{N}_1(\mathcal{F}, \alpha, x^n)}{n}}\Big).$$

3. **Linear predictors on the ℓ_1-ball.** Let $\mathcal{X}$ be the unit ℓ_1-ball in $\mathbb{R}^d$, and define

$$\mathcal{F} = \{f_w(x) = \langle w, x \rangle : \; \|w\|_\infty \le 1\}.$$

Show that the Rademacher complexity satisfies

$$\widehat{\mathfrak{R}}(\mathcal{F}) = \mathcal{O}\Big(\sqrt{\frac{d \log n}{n}}\Big).$$

4. **Chaining and Dudley's integral.** Using chaining and Dudley's inequality, show that

$$\widehat{\mathfrak{R}}(\mathcal{F}) \;\le\; \inf_{\alpha \ge 0} \Big(4\alpha + \frac{12}{\sqrt{n}} \int_\alpha^1 \sqrt{\log \mathcal{N}_2(\mathcal{F}, \delta, x^n)}\, d\delta\Big).$$

5. **Monotone functions and covering bounds.** Let $\mathcal{F}$ be a class of monotone (non-decreasing) functions on $\mathbb{R}$. Show that

$$\mathcal{N}_\infty(\mathcal{F}, \alpha, x^n) \;\le\; \Big(\frac{n}{\alpha}\Big)^2.$$

Compare the generalization bounds obtained in the previous parts when specialized to this class.

6. **Intuition behind fat-shattering.** Give an intuitive explanation of the fat-shattering dimension and how it generalizes the VC dimension to real-valued function classes.

7. **Fat-shattering of monotone functions.** For the class of monotone functions on $\mathbb{R}$, show that

$$\mathrm{Fatdim}(\mathcal{F}, \alpha) \;\le\; \frac{4}{\alpha}.$$

8. **Covering numbers via fat-shattering.** Show that for any function class $\mathcal{F} \subseteq \{\mathcal{X} \to [-1,1]\}$,

$$\mathcal{N}_2(\mathcal{F}, \alpha, x^n) \;\le\; \Big(\frac{2}{\alpha}\Big)^{K \cdot \mathrm{Fatdim}(\mathcal{F}, c\alpha)},$$

where $K, c > 0$ are universal constants.

9. **Dudley bound in terms of fat-shattering.** Using the previous part, prove that

$$\widehat{\mathfrak{R}}(\mathcal{F}) \;\le\; \inf_{\alpha \ge 0} \Big\{ 4\alpha + \frac{12}{\sqrt{n}} \int_\alpha^1 \sqrt{K \cdot \mathrm{Fatdim}(\mathcal{F}, c\delta) \cdot \log\Big(\frac{2}{\delta}\Big)}\, d\delta \Big\},$$

where $K, c > 0$ are universal constants.

Solution

Throughout, identify $\mathcal{F}$ with its evaluation set on the sample

$$\mathcal{T} := \{(f(x_1), \dots, f(x_n)) : \; f \in \mathcal{F}\} \subseteq [-1,1]^n,$$

and use the empirical metrics $d_p(u,v) := \Big(\frac{1}{n}\sum_{i=1}^n |u_i - v_i|^p\Big)^{1/p}$ (with the usual $p = \infty$ convention). Then $\mathcal{N}_p(\mathcal{F}, \alpha, x^n) = \mathcal{N}(\mathcal{T}, d_p, \alpha)$.

(1) Metric monotonicity. For $1 \le p \le q \le \infty$, norm monotonicity on $\mathbb{R}^n$ gives $d_p(u,v) \le d_q(u,v)$ for all u, v. Hence any α-net for $(\mathcal{T}, d_q)$ is also an α-net for $(\mathcal{T}, d_p)$, so $\mathcal{N}_p(\mathcal{F}, \alpha, x^n) \le \mathcal{N}_q(\mathcal{F}, \alpha, x^n)$.

(2) Rademacher via ℓ_1-covering. Let $\varepsilon_1, \dots, \varepsilon_n$ be i.i.d. Rademachers and

$$\widehat{\mathfrak{R}}(\mathcal{F}) := \mathbb{E}_\varepsilon\Big[\sup_{a \in \mathcal{T}} \frac{1}{n}\sum_{i=1}^n \varepsilon_i a_i\Big].$$

Fix $\alpha \ge 0$ and let $\mathcal{C} \subset \mathbb{R}^n$ be a minimal α-net of $(\mathcal{T}, d_1)$, so $|\mathcal{C}| = \mathcal{N}_1(\mathcal{F}, \alpha, x^n)$. For each $a \in \mathcal{T}$ pick $c(a) \in \mathcal{C}$ with $d_1(a, c(a)) \le \alpha$. Then

$$\frac{1}{n}\sum_{i=1}^n \varepsilon_i a_i = \frac{1}{n}\sum_{i=1}^n \varepsilon_i (a_i - c(a)_i) \;+\; \frac{1}{n}\sum_{i=1}^n \varepsilon_i c(a)_i,$$

and the first term is bounded by $d_1(a, c(a)) \le \alpha$ (since $|\varepsilon_i| = 1$). Taking $\sup_a$ and expectation:

$$\widehat{\mathfrak{R}}(\mathcal{F}) \le \alpha + \mathbb{E}_\varepsilon\Big[\sup_{c \in \mathcal{C}} \frac{1}{n}\sum_{i=1}^n \varepsilon_i c_i\Big].$$

Now use Massart's lemma: for a finite set $\mathcal{C}$ with $\max_{c \in \mathcal{C}} \|c\|_2 \le r$,

$$\mathbb{E}_\varepsilon\Big[\sup_{c \in \mathcal{C}} \frac{1}{n}\sum_{i=1}^n \varepsilon_i c_i\Big] \le \frac{r}{n}\sqrt{2 \log |\mathcal{C}|}.$$

Here $c \in [-1,1]^n$ so $\|c\|_2 \le \sqrt{n}$, hence

$$\widehat{\mathfrak{R}}(\mathcal{F}) \le \alpha + \sqrt{\frac{2\log \mathcal{N}_1(\mathcal{F},\alpha,x^n)}{n}}.$$

Take $\inf_{\alpha\ge 0}$.

(3) Linear predictors on the ℓ_1-ball. Let $\mathcal{X} = \{x \in \mathbb{R}^d : \|x\|_1 \le 1\}$ and $\mathcal{F} = \{x \mapsto \langle w, x\rangle : \ \|w\|_\infty \le 1\}$. Then

$$\widehat{\mathfrak{R}}(\mathcal{F}) = \mathbb{E}_\varepsilon\Big[\sup_{\|w\|_\infty\le 1}\Big\langle w, \frac{1}{n}\sum_{i=1}^n \varepsilon_i x_i\Big\rangle\Big] = \frac{1}{n}\,\mathbb{E}_\varepsilon\Big[\Big\|\sum_{i=1}^n \varepsilon_i x_i\Big\|_1\Big].$$

Write $x_i = (x_{i,1},\dots,x_{i,d})$. By Khintchine and Cauchy–Schwarz,

$$\mathbb{E}\Big[\Big\|\sum_{i=1}^n \varepsilon_i x_i\Big\|_1\Big] = \sum_{j=1}^d \mathbb{E}\Big[\Big|\sum_{i=1}^n \varepsilon_i x_{i,j}\Big|\Big] \le \sum_{j=1}^d \Big(\sum_{i=1}^n x_{i,j}^2\Big)^{1/2} \le \sqrt{d}\Big(\sum_{i=1}^n \|x_i\|_2^2\Big)^{1/2}.$$

Since $\|x_i\|_2 \le \|x_i\|_1 \le 1$, we get $\sum_i \|x_i\|_2^2 \le n$, hence

$$\widehat{\mathfrak{R}}(\mathcal{F}) \le \sqrt{\frac{d}{n}} = O\Big(\sqrt{\frac{d\log n}{n}}\Big).$$

(4) Chaining / Dudley. Consider the process $X_a := \frac{1}{n}\sum_{i=1}^n \varepsilon_i a_i$ indexed by $a \in \mathcal{T}$. For $a, b \in \mathcal{T}$, Hoeffding (or standard Rademacher mgf) gives sub-Gaussian increments:

$$\mathbb{E}_\varepsilon \exp\Big(\lambda(X_a - X_b)\Big) \le \exp\Big(\frac{\lambda^2}{2n}\, d_2(a,b)^2\Big).$$

Dudley's bound for sub-Gaussian processes yields

$$\widehat{\mathfrak{R}}(\mathcal{F}) = \mathbb{E}[\sup_{a\in\mathcal{T}} X_a] \le \inf_{\alpha\ge 0}\Big(4\alpha + \frac{12}{\sqrt{n}}\int_\alpha^1 \sqrt{\log \mathcal{N}(\mathcal{T}, d_2, \delta)}\, d\delta\Big),$$

i.e.

$$\widehat{\mathfrak{R}}(\mathcal{F}) \le \inf_{\alpha\ge 0}\Big(4\alpha + \frac{12}{\sqrt{n}}\int_\alpha^1 \sqrt{\log \mathcal{N}_2(\mathcal{F}, \delta, x^n)}\, d\delta\Big).$$

(5) Monotone class: $\mathcal{N}_\infty$ bound and comparison. Let $\mathcal{F}$ be monotone non-decreasing functions on $\mathbb{R}$. Sort the sample so $x_1 \le \cdots \le x_n$. Then every $a \in \mathcal{T}$ is non-decreasing. Quantize $[-1,1]$ with a grid of step α, of size $m \le 2/\alpha$. A non-decreasing length-n sequence over m levels is determined by counts (stars-and-bars), so

$$\mathcal{N}_\infty(\mathcal{F},\alpha,x^n) \ \le\ \binom{n+m}{m} \ \le\ (n+m)^m \ \le\ n^{2/\alpha}.$$

Using (1), $\mathcal{N}_1, \mathcal{N}_2 \le \mathcal{N}_\infty$. Plugging into (2) gives a crude rate $\widehat{\mathfrak{R}}(\mathcal{F}) \lesssim (\log n/n)^{1/3}$, while plugging into (4) gives $\widehat{\mathfrak{R}}(\mathcal{F}) \lesssim \sqrt{\log n/n}$ (sharper).

(6) Fat-shattering intuition. VC dimension counts how many points can be labeled by

thresholds for *binary* outputs. Fat-shattering does the same for *real* outputs but enforces a margin: points labeled 1 must sit above a threshold by $\alpha/2$ and points labeled 0 below it by $\alpha/2$. Larger α means fewer points can be separated.

(7) Fatdim of monotone functions. If $\{x_1 < \cdots < x_m\}$ is α-shattered with thresholds $y_i \in [-1, 1]$, then by pigeonhole there exist $i > j$ with $|y_i - y_j| < \alpha$ once $m > 4/\alpha$ (range length 2). Choose the labeling that forces $f(x_j) > y_j + \alpha/2$ and $f(x_i) < y_i - \alpha/2$; then $f(x_j) > f(x_i)$ contradicts monotonicity. Hence $\mathrm{Fatdim}(\mathcal{F}, \alpha) \leq 4/\alpha$.

(8) Covering via fat-shattering. A standard entropy bound for bounded classes gives: there exist universal $K, c > 0$ such that for any sample,

$$\mathcal{N}_2(\mathcal{F}, \alpha, x^n) \;\leq\; \Big(\frac{2}{\alpha}\Big)^{K\,\mathrm{Fatdim}(\mathcal{F}, c\alpha)}.$$

(9) Dudley in terms of fat-shattering. Combine (4) with (8): for $\delta \in (0, 1]$,

$$\log \mathcal{N}_2(\mathcal{F}, \delta, x^n) \leq K\,\mathrm{Fatdim}(\mathcal{F}, c\delta)\,\log\Big(\frac{2}{\delta}\Big),$$

so

$$\widehat{\mathfrak{R}}(\mathcal{F}) \leq \inf_{\alpha \geq 0} \Big\{4\alpha + \frac{12}{\sqrt{n}} \int_\alpha^1 \sqrt{K\,\mathrm{Fatdim}(\mathcal{F}, c\delta)\,\log\Big(\frac{2}{\delta}\Big)}\,d\delta\Big\}.$$

Problem 3

Natarajan Dimension. Let $\mathcal{H} \subseteq \{h : \mathcal{X} \to [k]\}$ where $[k] = \{1, \ldots, k\}$. A set $C \subseteq \mathcal{X}$ is *Natarajan-shattered* by $\mathcal{H}$ if there exist $f_0, f_1 : C \to [k]$ such that (i) $f_0(x) \neq f_1(x)$ for all $x \in C$, and (ii) for every $B \subseteq C$ there is $h \in \mathcal{H}$ with $h|_B = f_0|_B$ and $h|_{C \setminus B} = f_1|_{C \setminus B}$. Define $\mathrm{Ndim}(\mathcal{H})$ as the maximum size of a shattered set.

(1) Show that for $k = 2$, $\mathrm{Ndim}(\mathcal{H}) = \mathrm{VCdim}(\mathcal{H})$.

(2) Show the growth bound

$$|\mathcal{H}| \;\leq\; |\mathcal{X}|^{\mathrm{Ndim}(\mathcal{H})} \cdot k^{2\,\mathrm{Ndim}(\mathcal{H})}.$$

Solution

(1) Binary reduction ($k = 2$). Let labels be $\{0, 1\}$. If C is VC-shattered, take $f_0 \equiv 0$ and $f_1 \equiv 1$ on C; then the VC property gives the required h for every $B \subseteq C$, hence C is Natarajan-shattered and Ndim $\geq$ VCdim. Conversely, if C is Natarajan-shattered by f_0, f_1, then for each $x \in C$ we have $\{f_0(x), f_1(x)\} = \{0, 1\}$. Given any $B \subseteq C$, property (ii) yields h with $h(x) = f_0(x)$ on B and $h(x) = f_1(x)$ on $C \setminus B$, which realizes every binary labeling of C. Thus C is VC-shattered and VCdim $\geq$ Ndim. Therefore $\mathrm{Ndim}(\mathcal{H}) = \mathrm{VCdim}(\mathcal{H})$ when $k = 2$.

(2) Growth bound. Let $n := |\mathcal{X}|$ and $d := \mathrm{Ndim}(\mathcal{H})$. Define

$$S(n, d) := \max\{|\mathcal{G}| : \; \mathcal{G} \subseteq [k]^{\mathcal{X}}, \; |\mathcal{X}| = n, \; \mathrm{Ndim}(\mathcal{G}) \leq d\}.$$

We show the recurrence $S(n, d) \leq S(n-1, d) + \binom{k}{2} S(n-1, d-1)$.
Fix $\mathcal{X} = \mathcal{X}' \cup \{x\}$ with $|\mathcal{X}'| = n - 1$ and a class $\mathcal{G} \subseteq [k]^{\mathcal{X}}$ with $\mathrm{Ndim}(\mathcal{G}) \leq d$. Let $\mathcal{G}' := \{g|_{\mathcal{X}'} : \; g \in \mathcal{G}\} \subseteq [k]^{\mathcal{X}'}$. For each restriction $g' \in \mathcal{G}'$ define the set of attainable labels at x:

$$T(g') := \{\ell \in [k] : \exists g \in \mathcal{G} \text{ s.t. } g|_{\mathcal{X}'} = g', \; g(x) = \ell\}.$$

Partition $\mathcal{G}'$ by these sets: $\mathcal{G}_T := \{g' \in \mathcal{G}' : T(g') = T\}$. Each $g' \in \mathcal{G}_T$ has exactly $|T|$ extensions in $\mathcal{G}$, hence

$$|\mathcal{G}| = \sum_{T \subseteq [k]} |T| \, |\mathcal{G}_T| = |\mathcal{G}'| + \sum_{T : |T| \geq 2} (|T| - 1) \, |\mathcal{G}_T|. \tag{*}$$

For each unordered pair $\{i, j\} \subseteq [k]$ define

$$\mathcal{G}_{ij} := \{g' \in \mathcal{G}' : \; i, j \in T(g')\}.$$

Then $\sum_{i<j} |\mathcal{G}_{ij}| = \sum_T \binom{|T|}{2} |\mathcal{G}_T|$. Also $\binom{|T|}{2} \geq |T| - 1$ for $|T| \geq 1$, so

$$\sum_T (|T| - 1) \, |\mathcal{G}_T| \leq \sum_{i<j} |\mathcal{G}_{ij}|. \tag{**}$$

Key claim: for every $i \neq j$, $\mathrm{Ndim}(\mathcal{G}_{ij}) \leq d - 1$, hence $|\mathcal{G}_{ij}| \leq S(n-1, d-1)$. Indeed, if $\mathcal{G}_{ij}$

Natarajan-shattered some $C \subseteq \mathcal{X}'$ of size d, witnessed by (f_0, f_1) on C, then using the fact that every $g' \in \mathcal{G}_{ij}$ has extensions taking i and j at x, one can extend witnesses by setting $\tilde{f}_0(x) = j$, $\tilde{f}_1(x) = i$ to obtain shattering of $C \cup \{x\}$ of size $d+1$ by $\mathcal{G}$, contradicting $\mathrm{Ndim}(\mathcal{G}) \leq d$.
Thus $\sum_{i<j} |\mathcal{G}_{ij}| \leq \binom{k}{2} S(n-1, d-1)$, and also $|\mathcal{G}'| \leq S(n-1, d)$ (restriction cannot increase Natarajan dimension). Plugging these into $(*)$ using $(**)$ gives

$$|\mathcal{G}| \leq S(n-1, d) + \binom{k}{2} S(n-1, d-1),$$

so taking maxima yields the recurrence for $S(n, d)$.
Now prove by induction on n that

$$S(n, d) \leq \binom{k}{2}^d \sum_{t=0}^{d} \binom{n}{t}.$$

The base $n = 0$ is trivial. The induction step follows from the recurrence plus Pascal's identity $\binom{n}{t} = \binom{n-1}{t} + \binom{n-1}{t-1}$.
Finally, using $\sum_{t=0}^{d} \binom{n}{t} \leq n^d + 1 \leq 2n^d$ and $\binom{k}{2} \leq k^2$, we get

$$|\mathcal{H}| \leq S(n, d) \leq (k^2)^d \cdot 2n^d \;\leq\; n^d\, k^{2d} \quad \text{(up to an absolute constant)},$$

which matches the requested form.

Problem 4

Matrix Bernstein with intrinsic-dimension dependence. Let $\{A_k\}_{k=1}^n$ be independent random Hermitian matrices with $\mathbb{E}A_k = 0$ and $\|A_k\|_{\mathrm{op}} \le 1$. Define

$$S_n := \sum_{k=1}^n A_k, \qquad \phi(t) := e^t - t - 1, \qquad V := \sum_{k=1}^n \mathrm{Var}(A_k), \quad \mathrm{Var}(A) := \mathbb{E}[A^2].$$

(1) Show that for any $t > 0$ and $\delta > 0$,

$$\mathbb{P}(\lambda_{\max}(S_n) > \delta) \;\le\; \frac{\mathrm{Tr}(\mathbb{E}[\phi(tS_n)])}{\phi(t\delta)}.$$

(2) Verify that if $\|A\|_{\mathrm{op}} \le 1$ and $\mathbb{E}A = 0$, then for $t \ge 0$,

$$\log \mathbb{E}[e^{tA}] \;\preceq\; \phi(t)\,\mathrm{Var}(A).$$

(3) Prove the intrinsic-dimension Bernstein bound

$$\mathbb{P}(\lambda_{\max}(S_n) > \delta) \;\le\; \frac{\mathrm{Tr}(V)}{\|V\|_{\mathrm{op}}} \cdot \inf_{t>0} \left(\frac{\exp(\phi(t)\|V\|_{\mathrm{op}})}{\phi(t\delta)} \right).$$

(4) Let $X_1, \dots, X_n \in \mathbb{R}^d$ be i.i.d., $\mathbb{E}X_k = 0$, $\mathbb{E}[X_k X_k^\top] = C$, and $\|X_k\|_2 \le 1$. Use (3) to bound $\mathbb{P}(\|\sum_{k=1}^n X_k\|_2 > \delta)$. Also state the standard matrix Bernstein bound (dimension-dependent) for this event and compare.

Solution

(1) ϕ is increasing and $\phi(tS_n) \succeq 0$, so

$$\{\lambda_{\max}(S_n) > \delta\} \subseteq \{\lambda_{\max}(\phi(tS_n)) > \phi(t\delta)\} \subseteq \{\mathrm{Tr}(\phi(tS_n)) > \phi(t\delta)\}.$$

Markov on the nonnegative r.v. $\mathrm{Tr}(\phi(tS_n))$ gives the claim.

(2) For $\|A\|_{\mathrm{op}} \le 1$ one has the operator inequality

$$e^{tA} \;\preceq\; I + tA + \phi(t)A^2 \qquad (t \ge 0),$$

hence by $\mathbb{E}A = 0$,

$$\mathbb{E}e^{tA} \;\preceq\; I + \phi(t)\mathbb{E}[A^2].$$

Using operator monotonicity of log and $\log(I + X) \preceq X$ for $X \succeq 0$,

$$\log \mathbb{E}e^{tA} \;\preceq\; \log(I + \phi(t)\mathbb{E}[A^2]) \;\preceq\; \phi(t)\mathbb{E}[A^2].$$

(3) By the standard mgf argument for sums (Tropp/Lieb-style),

$$\log \mathbb{E}e^{tS_n} \;\preceq\; \sum_{k=1}^n \log \mathbb{E}e^{tA_k} \;\preceq\; \phi(t) \sum_{k=1}^n \mathbb{E}[A_k^2] \;=\; \phi(t)\,V,$$

so $\mathbb{E}e^{tS_n} \preceq e^{\phi(t)V}$ and therefore

$$\mathrm{Tr}\big(\mathbb{E}\phi(tS_n)\big) = \mathrm{Tr}\big(\mathbb{E}e^{tS_n}\big) - \mathrm{Tr}(I) \le \mathrm{Tr}\big(e^{\phi(t)V}\big) - d.$$

Let $\{\lambda_i\}_{i=1}^d$ be eigenvalues of V with $\lambda_1 = \|V\|_{\mathrm{op}}$. Then $\mathrm{Tr}(e^{\phi(t)V}) - d = \sum_i (e^{\phi(t)\lambda_i} - 1)$, and for $0 \le a \le b$, $e^a - 1 \le \frac{a}{b} e^b$. Applying with $a = \phi(t)\lambda_i$, $b = \phi(t)\lambda_1$ yields

$$e^{\phi(t)\lambda_i} - 1 \;\le\; \frac{\lambda_i}{\lambda_1}\, e^{\phi(t)\lambda_1}.$$

Summing gives $\mathrm{Tr}(\mathbb{E}\phi(tS_n)) \le \frac{\mathrm{Tr}(V)}{\|V\|_{\mathrm{op}}} e^{\phi(t)\|V\|_{\mathrm{op}}}$. Combine with (1) and take $\inf_{t>0}$.

(4) Use Hermitian dilation: $A_k = \begin{pmatrix} 0 & X_k^\top \\ X_k & 0 \end{pmatrix}$. Then $\|A_k\|_{\mathrm{op}} = \|X_k\|_2 \le 1$ and

$$\Big\| \sum_{k=1}^n X_k \Big\|_2 \;=\; \Big\| \sum_{k=1}^n A_k \Big\|_{\mathrm{op}}.$$

Also $A_k^2 = \mathrm{diag}(\|X_k\|_2^2,\ X_k X_k^\top)$, hence

$$V = \sum_{k=1}^n \mathbb{E}[A_k^2] = n \cdot \mathrm{diag}(\mathrm{Tr}(C),\ C).$$

So $\mathrm{Tr}(V) = 2n\,\mathrm{Tr}(C)$ and $\|V\|_{\mathrm{op}} = n\,\mathrm{Tr}(C)$, hence $\mathrm{Tr}(V)/\|V\|_{\mathrm{op}} = 2$. Finally, $\|S_n\|_{\mathrm{op}} > \delta$ implies $\lambda_{\max}(S_n) > \delta$ or $\lambda_{\max}(-S_n) > \delta$, so applying (3) to both gives

$$\mathbb{P}\Big(\Big\| \sum_{k=1}^n X_k \Big\|_2 > \delta\Big) \;\le\; 4 \cdot \inf_{t>0} \left(\frac{\exp\big(\phi(t)\, n\mathrm{Tr}(C)\big)}{\phi(t\delta)} \right).$$

In particular (Bernstein form), this implies for a universal $c > 0$,

$$\mathbb{P}\Big(\Big\| \sum_{k=1}^n X_k \Big\|_2 > \delta\Big) \;\le\; 4\exp\Big(-c \min\Big\{ \frac{\delta^2}{n\,\mathrm{Tr}(C)},\ \delta \Big\}\Big).$$

Standard matrix Bernstein comparison: applying classical matrix Bernstein to the same dilation gives a prefactor $(d+1)$ (ambient dimension),

$$\mathbb{P}\Big(\Big\| \sum_{k=1}^n X_k \Big\|_2 > \delta\Big) \;\lesssim\; (d+1)\exp\Big(-c \min\Big\{ \frac{\delta^2}{n\|C\|_{\mathrm{op}}},\ \delta \Big\}\Big).$$

Advantage: the intrinsic bound replaces ambient dimension $(d+1)$ by the **effective rank** $r(V) := \mathrm{Tr}(V)/\|V\|_{\mathrm{op}}$ (here $r(V) = 2$), which can be *much smaller* than d.

Problem 5

Matrix Moment Generating Functions. Let X be a random Hermitian matrix. Define the matrix MGF and CGF by

$$M_X(\theta) := \mathbb{E}[e^{\theta X}], \qquad \Xi_X(\theta) := \log \mathbb{E}[e^{\theta X}].$$

(1) (**Trace MGF tail bounds**) Let Y be a random Hermitian matrix. Show that for every $t \in \mathbb{R}$,

$$\mathbb{P}\{\lambda_{\max}(Y) \geq t\} \leq \inf_{\theta>0} \left(e^{-\theta t}\, \mathbb{E}[\mathrm{Tr}(e^{\theta Y})]\right), \qquad \mathbb{P}\{\lambda_{\min}(Y) \leq t\} \leq \inf_{\theta<0} \left(e^{-\theta t}\, \mathbb{E}[\mathrm{Tr}(e^{\theta Y})]\right).$$

(2) (**Bounds on expected extreme eigenvalues**) Show that

$$\mathbb{E}[\lambda_{\max}(Y)] \leq \inf_{\theta>0} \left(\frac{1}{\theta} \log \mathbb{E}[\mathrm{Tr}(e^{\theta Y})]\right), \qquad \mathbb{E}[\lambda_{\min}(Y)] \geq \sup_{\theta<0} \left(\frac{1}{\theta} \log \mathbb{E}[\mathrm{Tr}(e^{\theta Y})]\right).$$

(3) (**Failure of MGF multiplicativity**) Show that even if $X_1, \ldots, X_n$ are independent Hermitian random matrices,

$$M_{\sum_{k=1}^n X_k}(\theta) \neq \prod_{k=1}^n M_{X_k}(\theta) \quad \text{in general.}$$

(4) (**Failure of trace exponential factorization**) Show that there exist Hermitian matrices A, H, T such that

$$\mathrm{Tr}(e^{A+H+T}) \not\leq \mathrm{Tr}(e^A e^H e^T).$$

(5) (**Failure of CGF additivity**) Show that even for independent Hermitian random matrices,

$$\Xi_{\sum_{k=1}^n X_k}(\theta) \neq \sum_{k=1}^n \Xi_{X_k}(\theta) \quad \text{in general.}$$

Solution

(1) Fix $\theta > 0$. On $\{\lambda_{\max}(Y) \geq t\}$ we have $e^{\theta \lambda_{\max}(Y)} \geq e^{\theta t}$, i.e.

$$\lambda_{\max}(e^{\theta Y}) = e^{\theta \lambda_{\max}(Y)} \geq e^{\theta t}.$$

Since $e^{\theta Y} \succeq 0$, $\lambda_{\max}(e^{\theta Y}) \leq \mathrm{Tr}(e^{\theta Y})$, hence

$$\{\lambda_{\max}(Y) \geq t\} \subseteq \{\mathrm{Tr}(e^{\theta Y}) \geq e^{\theta t}\}.$$

Markov yields

$$\mathbb{P}\{\lambda_{\max}(Y) \geq t\} \leq e^{-\theta t}\, \mathbb{E}[\mathrm{Tr}(e^{\theta Y})].$$

Take $\inf_{\theta>0}$. For $\lambda_{\min}$, fix $\theta < 0$; the map $u \mapsto e^{\theta u}$ is decreasing, so on $\{\lambda_{\min}(Y) \leq t\}$

we have $e^{\theta\lambda_{\min}(Y)} \geq e^{\theta t}$ and $\lambda_{\max}(e^{\theta Y}) = e^{\theta\lambda_{\min}(Y)} \geq e^{\theta t}$. Repeat the same trace/Markov argument and take $\inf_{\theta<0}$.

(2) For $\theta > 0$, the scalar inequality $\max_i a_i \leq \log \sum_i e^{a_i}$ applied to $a_i = \theta\lambda_i(Y)$ gives

$$\theta\lambda_{\max}(Y) \leq \log\Big(\sum_{i=1}^{d} e^{\theta\lambda_i(Y)}\Big) = \log \operatorname{Tr}(e^{\theta Y}),$$

so $\lambda_{\max}(Y) \leq \frac{1}{\theta}\log \operatorname{Tr}(e^{\theta Y})$ a.s. Take expectations and use Jensen (log concave):

$$\mathbb{E}[\lambda_{\max}(Y)] \leq \frac{1}{\theta}\,\mathbb{E}[\log \operatorname{Tr}(e^{\theta Y})] \leq \frac{1}{\theta}\log \mathbb{E}[\operatorname{Tr}(e^{\theta Y})].$$

Then take $\inf_{\theta>0}$. For $\lambda_{\min}$, take $\theta < 0$ so that $\theta\lambda_{\min}(Y) = \max_i \theta\lambda_i(Y)$ and the same steps give $\mathbb{E}[\lambda_{\min}(Y)] \geq \sup_{\theta<0} \frac{1}{\theta}\log \mathbb{E}[\operatorname{Tr}(e^{\theta Y})]$.

(3) It suffices to give $n = 2$. Take deterministic (hence independent) Hermitian matrices

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

which do not commute ($AH \neq HA$). Let $X_1 \equiv A$, $X_2 \equiv H$. Then

$$M_{X_1+X_2}(\theta) = e^{\theta(A+H)}, \qquad M_{X_1}(\theta)M_{X_2}(\theta) = e^{\theta A}e^{\theta H}.$$

The identity $e^{\theta(A+H)} = e^{\theta A}e^{\theta H}$ would force commutativity; indeed comparing the θ^2 terms in Taylor expansions gives a mismatch unless $AH = HA$. Hence $M_{X_1+X_2}(\theta) \neq M_{X_1}(\theta)M_{X_2}(\theta)$ for generic $\theta \neq 0$.

(4) (Concrete counterexample.) Work with the same 2×2 Pauli pair σ_x, σ_z:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

Set

$$A = \sigma_x + \sigma_z = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H = -\sigma_z = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad T = -\sigma_x + \sigma_z = \begin{pmatrix} 1 & -1 \\ -1 & -1 \end{pmatrix}.$$

Then $A + H + T = \sigma_z$, so

$$\operatorname{Tr}(e^{A+H+T}) = \operatorname{Tr}(e^{\sigma_z}) = e^1 + e^{-1} = 2\cosh(1).$$

A direct closed-form multiplication using $e^{x\sigma_x+z\sigma_z} = \cosh(r)I + \frac{\sinh(r)}{r}(x\sigma_x + z\sigma_z)$ with $r = \sqrt{x^2+z^2}$ yields

$$\operatorname{Tr}(e^A e^H e^T) = 2\Big(\cosh(\sqrt{2})^2\cosh(1) - \sqrt{2}\,\cosh(\sqrt{2})\sinh(\sqrt{2})\sinh(1)\Big) \approx 0.63,$$

whereas $2\cosh(1) \approx 3.086$. Hence $\operatorname{Tr}(e^{A+H+T}) > \operatorname{Tr}(e^A e^H e^T)$, so the claimed factorization

inequality fails.

(5) Use an explicit independent example where both sides are scalar multiples of I but differ. Let $\varepsilon_1, \varepsilon_2$ be independent Rademachers and take the same anti-commuting Hermitians A, H as in (3). Define

$$X := \varepsilon_1 A, \qquad Y := \varepsilon_2 H.$$

Since $A^2 = H^2 = I$,

$$\mathbb{E}[e^{\theta X}] = \cosh(\theta) I, \qquad \mathbb{E}[e^{\theta Y}] = \cosh(\theta) I,$$

so

$$\Xi_X(\theta) + \Xi_Y(\theta) = 2 \log(\cosh \theta)\, I.$$

But $(\varepsilon_1 A + \varepsilon_2 H)^2 = 2I$ (cross-terms cancel because $AH + HA = 0$), hence

$$\mathbb{E}[e^{\theta(X+Y)}] = \cosh(\sqrt{2}\,\theta)\, I \quad \Rightarrow \quad \Xi_{X+Y}(\theta) = \log(\cosh(\sqrt{2}\,\theta))\, I.$$

For example at $\theta = 1$, $\log(\cosh \sqrt{2}) \neq 2 \log(\cosh 1)$, hence $\Xi_{X+Y}(1) \neq \Xi_X(1) + \Xi_Y(1)$ and CGF additivity fails in general.

Problem 6

Matrix Bernstein inequality with unbounded summands. Let $\{A_i\}_{i=1}^n$ be independent random matrices in $\mathbb{R}^{d_1 \times d_2}$, and suppose

$$A_i = G_i\, B_i,$$

where G_i is a real random variable with $\mathbb{E}[G_i] = 0$, and B_i is a random matrix independent of G_i. Assume that for every integer $k \geq 3$,

$$\mathbb{E}\big[|G_i|^k\big] \leq \frac{k!}{2}\, b_1^{k-2}\, \sigma^2,$$

and that

$$\|B_i\|_{\mathrm{op}} \leq b_2 \quad \text{a.s.}$$

Show that for every $\delta > 0$,

$$\mathbb{P}\left[\left\|\frac{1}{n}\sum_{i=1}^n A_i\right\|_{\mathrm{op}} \geq \delta\right] \leq 2(d_1 + d_2)\exp\left(-\frac{n\delta^2}{2(\sigma^2 b_2^2 + b_1 b_2\, \delta)}\right).$$

Solution

Step 1 (Dilation: reduce rectangular to symmetric). Let $d := d_1 + d_2$ and define the self-adjoint dilation

$$X_i := \begin{pmatrix} 0 & A_i \\ A_i^\top & 0 \end{pmatrix} \in \mathbb{R}^{d \times d}, \qquad Y_i := \begin{pmatrix} 0 & B_i \\ B_i^\top & 0 \end{pmatrix}.$$

Then $X_i = G_i Y_i$ and X_i are independent symmetric matrices. Moreover,

$$\|X_i\|_{\mathrm{op}} = \|A_i\|_{\mathrm{op}}, \qquad \left\|\sum_{i=1}^n X_i\right\|_{\mathrm{op}} = \left\|\sum_{i=1}^n A_i\right\|_{\mathrm{op}}.$$

Also $\mathbb{E}X_i = 0$ because $\mathbb{E}[G_i] = 0$ and $G_i \perp B_i$.
So it suffices to bound

$$\mathbb{P}\left[\left\|\frac{1}{n}\sum_{i=1}^n X_i\right\|_{\mathrm{op}} \geq \delta\right].$$

Step 2 (Trace–Markov / Laplace transform). Let $S := \sum_{i=1}^n X_i$ (symmetric). For $\theta > 0$,

$$\mathbb{P}[\lambda_{\max}(S) \geq n\delta] \leq e^{-\theta n \delta}\, \mathbb{E}\,\mathrm{Tr}\, e^{\theta S},$$

since $e^{\theta \lambda_{\max}(S)} = \lambda_{\max}(e^{\theta S}) \leq \mathrm{Tr}(e^{\theta S})$.

Step 3 (Separate the sum via Lieb). By Lieb's trace mgf inequality (Theorem 1.14),

$$\mathbb{E}\,\mathrm{Tr}\, e^{\theta \sum_{i=1}^n X_i} \leq \mathrm{Tr}\exp\left(\sum_{i=1}^n \log \mathbb{E}\, e^{\theta X_i}\right).$$

So we need an upper bound on $\log \mathbb{E}e^{\theta X_i}$.

Step 4 (Bound each $\log \mathbb{E}e^{\theta X_i}$). Fix i. Condition on Y_i. Since Y_i is symmetric and independent of G_i, diagonalize $Y_i = U\Lambda U^\top$, $|\lambda_j| \le \|Y_i\|_{\mathrm{op}} = \|B_i\|_{\mathrm{op}} \le b_2$ a.s. Then

$$\mathbb{E}\Big[e^{\theta X_i} \mid Y_i\Big] = U \operatorname{diag}\Big(\mathbb{E}e^{\theta \lambda_j G_i}\Big)_j U^\top.$$

Use the scalar Bernstein-moment $\Rightarrow$ MGF bound (Lemma 1.15): for $|t| < 1/b_1$,

$$\mathbb{E}e^{tG_i} \le \exp\left(\frac{t^2\sigma^2}{2(1-b_1|t|)}\right).$$

Take $t = \theta\lambda_j$. If $0 < \theta < 1/(b_1 b_2)$ then $|t| \le \theta b_2 < 1/b_1$, and

$$\mathbb{E}e^{\theta\lambda_j G_i} \le \exp\left(\frac{\theta^2\lambda_j^2\sigma^2}{2(1-b_1b_2\theta)}\right).$$

Hence (as a matrix inequality)

$$\mathbb{E}\Big[e^{\theta X_i} \mid Y_i\Big] \preceq \exp\left(\frac{\theta^2\sigma^2}{2(1-b_1b_2\theta)}\, Y_i^2\right).$$

But $Y_i^2 \preceq \|Y_i\|_{\mathrm{op}}^2 I \preceq b_2^2 I$, so

$$\mathbb{E}e^{\theta X_i} = \mathbb{E}\,\mathbb{E}[e^{\theta X_i} \mid Y_i] \preceq \exp\left(\frac{\theta^2\sigma^2 b_2^2}{2(1-b_1b_2\theta)}\right) I.$$

Since log is operator-monotone on positive definite matrices,

$$\log \mathbb{E}e^{\theta X_i} \preceq \frac{\theta^2\sigma^2 b_2^2}{2(1-b_1b_2\theta)}\, I, \qquad 0 < \theta < \frac{1}{b_1 b_2}.$$

Step 5 (Trace bound + optimize θ). Summing over i and using $\operatorname{Tr}(e^{cI}) = d\, e^c$,

$$\mathbb{E}\operatorname{Tr} e^{\theta S} \le (d_1+d_2)\exp\left(\frac{n\theta^2\sigma^2 b_2^2}{2(1-b_1b_2\theta)}\right).$$

Therefore for $0 < \theta < 1/(b_1b_2)$,

$$\mathbb{P}\left[\lambda_{\max}\left(\frac{1}{n}S\right) \ge \delta\right] \le (d_1+d_2)\exp\left(-\theta n\delta + \frac{n\theta^2\sigma^2 b_2^2}{2(1-b_1b_2\theta)}\right).$$

Choose the standard Bernstein choice

$$\theta := \frac{\delta}{\sigma^2 b_2^2 + b_1 b_2\, \delta},$$

which satisfies $\theta < 1/(b_1 b_2)$. A direct substitution simplifies the exponent to

$$-\frac{n\delta^2}{2(\sigma^2 b_2^2 + b_1 b_2\,\delta)}.$$

So

$$\mathbb{P}\left[\lambda_{\max}\left(\frac{1}{n}S\right) \geq \delta\right] \leq (d_1 + d_2)\exp\left(-\frac{n\delta^2}{2(\sigma^2 b_2^2 + b_1 b_2\,\delta)}\right).$$

Step 6 (Two-sided + return to A_i). Apply the same bound to $-X_i$ and use $\|M\|_{\mathrm{op}} = \max\{\lambda_{\max}(M), \lambda_{\max}(-M)\}$ plus a union bound:

$$\mathbb{P}\left[\left\|\frac{1}{n}\sum_{i=1}^{n} X_i\right\|_{\mathrm{op}} \geq \delta\right] \leq 2(d_1 + d_2)\exp\left(-\frac{n\delta^2}{2(\sigma^2 b_2^2 + b_1 b_2\,\delta)}\right).$$

Finally, by dilation, $\left\|\frac{1}{n}\sum_{i=1}^{n} X_i\right\|_{\mathrm{op}} = \left\|\frac{1}{n}\sum_{i=1}^{n} A_i\right\|_{\mathrm{op}}$, which gives the desired inequality.

Problem 7

Matrix Khintchine inequality. Let $\varepsilon_1, \dots, \varepsilon_n$ be independent Rademacher random variables, and let $A_1, \dots, A_n$ be deterministic symmetric matrices in $\mathbb{R}^{d\times d}$. Show that for every $p \in [1, \infty)$,

$$\Big(\mathbb{E}\Big\|\sum_{i=1}^n \varepsilon_i A_i\Big\|_{\mathrm{op}}^p\Big)^{1/p} \;\le\; C\,\sqrt{p+\log d}\,\Big\|\sum_{i=1}^n A_i^2\Big\|_{\mathrm{op}}^{1/2},$$

for a universal constant $C > 0$.

Solution

Let

$$S := \sum_{i=1}^n \varepsilon_i A_i, \qquad V := \sum_{i=1}^n A_i^2 \succeq 0, \qquad \sigma^2 := \|V\|_{\mathrm{op}}.$$

Step 1 (Laplace transform bound for $\lambda_{\max}$). For any $\theta > 0$ and $t > 0$,

$$\mathbb{P}(\lambda_{\max}(S) \ge t) = \mathbb{P}(e^{\theta\lambda_{\max}(S)} \ge e^{\theta t}) \le e^{-\theta t}\,\mathbb{E}\,e^{\theta\lambda_{\max}(S)}.$$

Since S is symmetric, $e^{\theta S} \succ 0$ and $e^{\theta\lambda_{\max}(S)} = \lambda_{\max}(e^{\theta S}) \le \mathrm{Tr}(e^{\theta S})$, so

$$\mathbb{P}(\lambda_{\max}(S) \ge t) \le e^{-\theta t}\,\mathbb{E}\,\mathrm{Tr}(e^{\theta S}). \tag{1}$$

Step 2 (Matrix MGF bound: $\mathbb{E}\mathrm{Tr}(e^{\theta S}) \le \mathrm{Tr}(e^{\theta^2 V/2})$). Define partial sums $S_k := \sum_{i=1}^k \varepsilon_i A_i$ with $S_0 = 0$. Condition on $\varepsilon_1, \dots, \varepsilon_{k-1}$ so that S_{k-1} is deterministic. Then

$$\mathbb{E}_{\varepsilon_k}\big[e^{\theta S_k} \,\big|\, \varepsilon_{1:k-1}\big] = \frac{1}{2}\Big(e^{\theta(S_{k-1}+A_k)} + e^{\theta(S_{k-1}-A_k)}\Big).$$

Taking trace and using Golden–Thompson,

$$\mathrm{Tr}\,e^{\theta(S_{k-1}\pm A_k)} \le \mathrm{Tr}\big(e^{\theta S_{k-1}} e^{\pm\theta A_k}\big),$$

hence

$$\mathrm{Tr}\,\mathbb{E}_{\varepsilon_k}\big[e^{\theta S_k} \,\big|\, \varepsilon_{1:k-1}\big] \le \mathrm{Tr}\Big(e^{\theta S_{k-1}} \cdot \frac{e^{\theta A_k} + e^{-\theta A_k}}{2}\Big).$$

By the matrix inequality (Lemma 1.19),

$$\frac{e^B + e^{-B}}{2} \preceq e^{B^2/2} \quad \text{for symmetric } B,$$

so with $B = \theta A_k$ we obtain

$$\frac{e^{\theta A_k} + e^{-\theta A_k}}{2} \preceq e^{\theta^2 A_k^2/2}.$$

Therefore,

$$\mathrm{Tr}\,\mathbb{E}_{\varepsilon_k}\big[e^{\theta S_k} \,\big|\, \varepsilon_{1:k-1}\big] \le \mathrm{Tr}\big(e^{\theta S_{k-1}} e^{\theta^2 A_k^2/2}\big).$$

Iterating this inequality for $k = 1, \dots, n$ (peeling off $\varepsilon_n, \varepsilon_{n-1}, \dots$) yields

$$\mathbb{E}\,\mathrm{Tr}(e^{\theta S}) \le \mathrm{Tr}\exp\Big(\frac{\theta^2}{2}\sum_{i=1}^{n} A_i^2\Big) = \mathrm{Tr}\big(e^{\theta^2 V/2}\big). \tag{2}$$

Step 3 (Subgaussian tail for $\|S\|_{\mathrm{op}}$). Since $V \succeq 0$,

$$\mathrm{Tr}\big(e^{\theta^2 V/2}\big) = \sum_{j=1}^{d} e^{\theta^2 \lambda_j(V)/2} \le d\, e^{\theta^2 \|V\|_{\mathrm{op}}/2} = d\, e^{\theta^2 \sigma^2/2}.$$

Plugging into (1)–(2) gives

$$\mathbb{P}(\lambda_{\max}(S) \ge t) \le d \exp\Big(-\theta t + \frac{\theta^2\sigma^2}{2}\Big).$$

Optimizing over $\theta > 0$ (take $\theta = t/\sigma^2$) yields

$$\mathbb{P}(\lambda_{\max}(S) \ge t) \le d \exp\Big(-\frac{t^2}{2\sigma^2}\Big).$$

Apply the same bound to $-S$ (since $-S = \sum \varepsilon_i(-A_i)$ and $\sum(-A_i)^2 = V$), then union bound:

$$\mathbb{P}(\|S\|_{\mathrm{op}} \ge t) \le 2d \exp\Big(-\frac{t^2}{2\sigma^2}\Big). \tag{3}$$

Step 4 (Convert tail to moments). For $Z \ge 0$ and $p > 0$,

$$\mathbb{E}[Z^p] = \int_0^\infty p\, t^{p-1}\, \mathbb{P}(Z \ge t)\, dt.$$

With $Z = \|S\|_{\mathrm{op}}$ and (3),

$$\mathbb{E}\|S\|_{\mathrm{op}}^p \le 2d\, p \int_0^\infty t^{p-1} e^{-t^2/(2\sigma^2)} dt.$$

Substitute $t = \sigma u$ to get

$$\mathbb{E}\|S\|_{\mathrm{op}}^p \le 2d\, p\, \sigma^p \int_0^\infty u^{p-1} e^{-u^2/2} du = 2d\, p\, \sigma^p\, I_p,$$

where $I_p := \int_0^\infty u^{p-1} e^{-u^2/2} du$ (Lemma 1.17). Using the Gamma form $I_p = 2^{p/2-1}\Gamma(p/2)$ and Stirling's bound for Γ gives

$$I_p^{1/p} \;\le\; c\sqrt{p} \quad \text{for a universal } c > 0 \text{ and all } p \ge 1.$$

Therefore

$$\Big(\mathbb{E}\|S\|_{\mathrm{op}}^p\Big)^{1/p} \le \sigma\,(2dp)^{1/p}\, I_p^{1/p} \;\le\; C_0\, \sigma\, \sqrt{p}\, d^{1/p}, \qquad p \ge 1,$$

for a universal $C_0 > 0$.

Step 5 (Turn $d^{1/p}$ into $\sqrt{p+\log d}$). For any $q \geq p$, monotonicity of L_r norms gives

$$\Big(\mathbb{E}\|S\|_{\mathrm{op}}^{p}\Big)^{1/p} \leq \Big(\mathbb{E}\|S\|_{\mathrm{op}}^{q}\Big)^{1/q} \leq C_0\,\sigma\,\sqrt{q}\,d^{1/q}.$$

Choose $q := p + \log d$. Then $d^{1/q} = \exp((\log d)/q) \leq e$, hence

$$\Big(\mathbb{E}\|S\|_{\mathrm{op}}^{p}\Big)^{1/p} \leq C\,\sigma\,\sqrt{p+\log d},$$

for a universal constant $C > 0$ (absorbing e and C_0). Recalling $\sigma^2 = \|V\|_{\mathrm{op}}$ completes the proof:

$$\Big(\mathbb{E}\Big\|\sum_{i=1}^{n}\varepsilon_i A_i\Big\|_{\mathrm{op}}^{p}\Big)^{1/p} \leq C\,\sqrt{p+\log d}\,\Big\|\sum_{i=1}^{n}A_i^2\Big\|_{\mathrm{op}}^{1/2}.$$

Problem 8

Generalization Bounds via VC Dimension. Let G be a family of functions $g : \mathcal{Z} \to [0,1]$. Let $S = (z_1, \ldots, z_m)$ be m i.i.d. samples from a distribution D over $\mathcal{Z}$.

1. Show that for every $g \in G$, with probability at least $1 - \delta$,

$$\mathbb{E}_{z\sim D}[g(z)] \;\le\; \frac{1}{m}\sum_{i=1}^{m} g(z_i) \;+\; 2R_m(G) \;+\; \sqrt{\frac{\log(1/\delta)}{2m}}.$$

2. Define

$$G := \{(x,y) \mapsto \mathbb{I}(h(x) \neq y) \;:\; h \in \mathcal{H}\},$$

where $\mathcal{H} \subset \{-1,+1\}^{\mathcal{X}}$ is a binary hypothesis class and $y \in \{-1,+1\}$. Show that

$$R_m(G) = \frac{1}{2}\, R_m(\mathcal{H}).$$

3. Define the **growth function** $\Pi_{\mathcal{H}}(m)$, explain its meaning, and using Massart's lemma show that

$$R_m(\mathcal{H}) \;\le\; \sqrt{\frac{2\log \Pi_{\mathcal{H}}(m)}{m}}.$$

4. Using the definition of VC dimension and Sauer's lemma, show that if $d := \mathrm{VCdim}(\mathcal{H})$, then for $m \ge d$,

$$\Pi_{\mathcal{H}}(m) \;\le\; \Big(\frac{em}{d}\Big)^{d}.$$

5. Combine the previous parts. Suppose $\mathrm{VCdim}(\mathcal{H}) = d$ and S is an i.i.d. sample of size m. Show that with probability at least $1-\delta$, every $h \in \mathcal{H}$ satisfies

$$R(h) \;\le\; \widehat{R}_S(h) \;+\; C_1\sqrt{\frac{\log(m/d)}{m/d}} \;+\; C_2\sqrt{\frac{\log(1/\delta)}{m}},$$

for universal constants $C_1, C_2 > 0$.

Solution

Notation. Let $\sigma_1, \ldots, \sigma_m$ be i.i.d. Rademacher signs. Define the (expected) Rademacher complexity

$$R_m(G) \;:=\; \mathbb{E}_{S\sim D^m}\,\mathbb{E}_{\sigma}\Big[\sup_{g\in G}\frac{1}{m}\sum_{i=1}^{m}\sigma_i\, g(z_i)\Big].$$

1) Rademacher generalization bound for $G \subset [0,1]^{\mathcal{Z}}$

Define the random functional

$$\Phi(S) \;:=\; \sup_{h\in G}\Big(\mathbb{E}[h(z)] - \frac{1}{m}\sum_{i=1}^{m} h(z_i)\Big).$$

Then for any fixed $g \in G$,

$$\mathbb{E}[g(z)] - \frac{1}{m}\sum_{i=1}^{m} g(z_i) \;\le\; \Phi(S).$$

McDiarmid. If S and S' differ only in coordinate i, then for any $h \in G$,

$$\Big|\Phi(S) - \Phi(S')\Big| \le \sup_{h\in G} \frac{1}{m}|h(z_i) - h(z_i')| \le \frac{1}{m},$$

since $h(\cdot) \in [0,1]$. Hence McDiarmid's inequality implies that with probability at least $1-\delta$,

$$\Phi(S) \;\le\; \mathbb{E}[\Phi(S)] \;+\; \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Symmetrization. Let $S' = (z_1', \dots, z_m') \sim D^m$ be an independent ghost sample. Then

$$\mathbb{E}[\Phi(S)] = \mathbb{E}_{S,S'}\Big[\sup_{h\in G} \frac{1}{m}\sum_{i=1}^{m} \big(h(z_i') - h(z_i)\big)\Big].$$

By the swap-symmetry of pairs (z_i, z_i') and introducing Rademacher signs,

$$\mathbb{E}[\Phi(S)] = \mathbb{E}_{S,S',\sigma}\Big[\sup_{h\in G} \frac{1}{m}\sum_{i=1}^{m} \sigma_i\big(h(z_i') - h(z_i)\big)\Big].$$

Split the sum and use $\sup(A_h + B_h) \le \sup A_h + \sup B_h$:

$$\begin{aligned}\mathbb{E}[\Phi(S)] &\le \mathbb{E}_{S',\sigma}\Big[\sup_{h\in G} \frac{1}{m}\sum_{i=1}^{m} \sigma_i h(z_i')\Big] + \mathbb{E}_{S,\sigma}\Big[\sup_{h\in G} \frac{1}{m}\sum_{i=1}^{m} (-\sigma_i) h(z_i)\Big] \\ &= R_m(G) + R_m(G) = 2R_m(G),\end{aligned}$$

since $S \overset{d}{=} S'$ and $(-\sigma_i) \overset{d}{=} \sigma_i$.
Combining with McDiarmid, with probability at least $1-\delta$,

$$\forall g \in G: \quad \mathbb{E}[g(z)] - \frac{1}{m}\sum_{i=1}^{m} g(z_i) \le 2R_m(G) + \sqrt{\frac{\log(1/\delta)}{2m}},$$

which rearranges to the desired inequality.

2) For 0–1 loss: $R_m(G) = \frac{1}{2}R_m(\mathcal{H})$

Assume $h(x), y \in \{-1, +1\}$. Then

$$\mathbb{I}(h(x) \neq y) = \frac{1 - yh(x)}{2}.$$

Hence

$$
\begin{aligned}
R_m(G) &= \mathbb{E}_S\mathbb{E}_\sigma\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i\, \mathbb{I}(h(x_i)\neq y_i)\Big] \\
&= \mathbb{E}_S\mathbb{E}_\sigma\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i\, \frac{1-y_i h(x_i)}{2}\Big] \\
&= \frac{1}{2}\,\mathbb{E}_\sigma\Big[\frac{1}{m}\sum_{i=1}^m \sigma_i\Big] + \frac{1}{2}\,\mathbb{E}_S\mathbb{E}_\sigma\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m (-y_i\sigma_i)\, h(x_i)\Big].
\end{aligned}
$$

The first term is 0 because $\mathbb{E}[\sigma_i]=0$. Conditioned on S, the variables $\tilde{\sigma}_i := -y_i\sigma_i$ are still i.i.d. Rademachers, so

$$
R_m(G) = \frac{1}{2}\,\mathbb{E}_S\mathbb{E}_{\tilde{\sigma}}\Big[\sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \tilde{\sigma}_i\, h(x_i)\Big] = \frac{1}{2}\, R_m(\mathcal{H}).
$$

3) Growth function and Massart's lemma bound

Definition (growth function). For $\mathcal{H}\subset\{-1,+1\}^{\mathcal{X}}$ define

$$
\Pi_{\mathcal{H}}(m) := \max_{x_1,\dots,x_m\in\mathcal{X}} \Big|\{(h(x_1),\dots,h(x_m)) :\ h\in\mathcal{H}\}\Big|.
$$

It is the maximal number of distinct label patterns realizable by $\mathcal{H}$ on m points.
Fix a sample $S=(x_1,\dots,x_m)$. Consider the finite set

$$
\mathcal{A}_S := \{(h(x_1),\dots,h(x_m)) \in \mathbb{R}^m :\ h\in\mathcal{H}\}.
$$

Then $|\mathcal{A}_S| \le \Pi_{\mathcal{H}}(m)$ and every $a\in\mathcal{A}_S$ satisfies $\|a\|_2=\sqrt{m}$. By Massart's lemma,

$$
\mathbb{E}_\sigma\Big[\max_{a\in\mathcal{A}_S} \frac{1}{m}\sum_{i=1}^m \sigma_i a_i\Big] \le \sqrt{\frac{2\log|\mathcal{A}_S|}{m}} \le \sqrt{\frac{2\log\Pi_{\mathcal{H}}(m)}{m}}.
$$

But

$$
\max_{a\in\mathcal{A}_S} \frac{1}{m}\sum_{i=1}^m \sigma_i a_i = \sup_{h\in\mathcal{H}} \frac{1}{m}\sum_{i=1}^m \sigma_i h(x_i),
$$

so taking expectation over S yields

$$
R_m(\mathcal{H}) \ \le\ \sqrt{\frac{2\log\Pi_{\mathcal{H}}(m)}{m}}.
$$

4) Sauer's lemma $\Rightarrow \Pi_{\mathcal{H}}(m) \le (em/d)^d$

Let $d := \mathrm{VCdim}(\mathcal{H})$. Sauer's lemma gives for all m,

$$
\Pi_{\mathcal{H}}(m) \le \sum_{i=0}^d \binom{m}{i}.
$$

For $m \geq d \geq 1$, the binomial-sum bound implies

$$\sum_{i=0}^{d} \binom{m}{i} \leq \left(\frac{em}{d}\right)^d,$$

hence

$$\Pi_{\mathcal{H}}(m) \leq \left(\frac{em}{d}\right)^d \qquad (m \geq d).$$

5) Final bound (combine everything)

Let $g_h(x, y) = \mathbb{I}(h(x) \neq y)$. Then

$$R(h) = \mathbb{E}[g_h(x, y)], \qquad \widehat{R}_S(h) = \frac{1}{m} \sum_{i=1}^{m} g_h(x_i, y_i).$$

From Part (1), with probability at least $1 - \delta$,

$$\forall h \in \mathcal{H}: \quad R(h) \leq \widehat{R}_S(h) + 2R_m(G) + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Using Part (2), $2R_m(G) = R_m(\mathcal{H})$, thus

$$R(h) \leq \widehat{R}_S(h) + R_m(\mathcal{H}) + \sqrt{\frac{\log(1/\delta)}{2m}}.$$

Now apply Parts (3) and (4):

$$R_m(\mathcal{H}) \leq \sqrt{\frac{2 \log \Pi_{\mathcal{H}}(m)}{m}} \leq \sqrt{\frac{2}{m} \log \left(\frac{em}{d}\right)^d} = \sqrt{\frac{2d}{m} \log \left(\frac{em}{d}\right)}.$$

Write $r = m/d$. Then the complexity term equals $\sqrt{\frac{2}{r} \log(er)}$. In the common regime $r \geq e$ we have $\log(er) = 1 + \log r \leq 2 \log r$, so

$$\sqrt{\frac{2}{r} \log(er)} \leq 2\sqrt{\frac{\log r}{r}} = 2\sqrt{\frac{\log(m/d)}{m/d}}.$$

Also

$$\sqrt{\frac{\log(1/\delta)}{2m}} = \frac{1}{\sqrt{2}} \sqrt{\frac{\log(1/\delta)}{m}}.$$

Therefore, for $m \geq ed$, with probability at least $1 - \delta$, for every $h \in \mathcal{H}$,

$$R(h) \leq \widehat{R}_S(h) + C_1 \sqrt{\frac{\log(m/d)}{m/d}} + C_2 \sqrt{\frac{\log(1/\delta)}{m}},$$

where one valid choice of universal constants is $C_1 = 2$ and $C_2 = 1/\sqrt{2}$.